

Baby Steps for Easier Accessibility

Our practice of making Alibaba Cloud Console more inclusive by taking baby steps

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ABSTRACT

Web content accessibility is essential for a product to serve a wider range of users and make content accessible for all users under different conditions. However, it is rarely mentioned in ICT industry and also not being widely addressed among most of the companies in China. Advocating for accessibility in an existing complex product poses more challenges and demands more effort. As we think everyone's experience matters and it is imperative to make Alibaba Cloud console more inclusive, we propose a method that making the accessibility adoption process easier for all enterprise scaled products by taking incremental steps. Starting from getting stakeholders onboard with explicit data, our practice includes a way to design an adapted, simplified yet comprehensible version of accessibility design guidelines, which can make the implementation of accessibility more executable in Chinese enterprises, as well as how to design a set of level definition that could help to prioritize tasks based on the user composition of a product. The approach could be a helpful reference for other companies and teams in China.

CCS CONCEPTS

• Human-centered computing; • Accessibility Empirical studies in accessibility; • Accessibility design and evaluation methods;

KEYWORDS

Inclusive Design, Accessibility Implementation, Enterprises Practice

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1 INTRODUCTION

Web accessibility is essential for developers and organizations to create high-quality websites and web tools that don't exclude people from using their products and services [1]. It offers a product the ability to embrace a diverse set of users, including people with disabilities. Making a website more accessible can improve its overall usability and benefit all users.

In China, web accessibility is still rarely mentioned in the fast-growing ICT (information and communications technology) industry. This is to some extent due to the context we are situated, which includes, but is not limited to, the lack of corresponding law enforcement, education and the overall low employment rate of persons with disability (PWD). In US, Section 508 requires federal agencies to develop, procure, maintain and use ICT that is accessible to people with disabilities - regardless of whether or not they work for the federal government [2]. Article 54 of the Law of the People's Republic of China on the Protection of Disabled Persons mentions that the state and society shall research and develop information communication technologies and products suitable for the use of disabled persons. However, compared to Section 508, no actionable measures or enforcement were listed [3]. Moreover, according to the latest statistics available, 96.6% of employed PWDs in China are engaged in manual labor [4]. Though smart phone is now commonly used by most people in their daily life, only a small proportion of employed PWDs need to work with ICT on a daily basis, which, in return, leads to inadequate public awareness towards ICT accessibility or web accessibility. The concept and value of web accessibility are still vague to most people. And most of ICT companies in China haven't paid much attention to web accessibility yet.

In comparison, many oversea ICT companies already have well-established accessibility guidelines or design methodologies deeply implanted in their design systems. In particular, several cloud platforms are already providing highly accessible websites for users. By performing easy checks [5] on 6 cloud platforms including our own - Alibaba Cloud Console, we found that Alibaba Cloud Console is much harder to access compared to other cloud platforms, such as Google Cloud Platform and Azure Portal. As Alibaba Cloud is now serving global business with more than 3 million customers in 200+ countries and regions, and actively expanding its global market. It is imperative that we take proactive measures on improving the accessibility of Alibaba Cloud Console that serves majority of our customers, so that we can provide better experience to a wide range of users in all conditions.

However, compared with creating a new accessible product from scratch, implementing accessibility in an existing product poses more challenges and demands more effort. In this paper, we will go

over some of the challenges we are facing and deep dive into how we overcome them to make Alibaba Cloud Console more accessible by taking baby steps.

2 CHALLENGES

Several working groups dedicated to improving accessibility already exist in Alibaba Inc., but we were still faced with a lot of push backs when we brought up the topic of improving accessibility of Alibaba Cloud Console. It's hard to advocate for accessibility in our cloud console given that people still think "apparently only an extremely small amount of our users is with disability", not to mention the high implementation cost accompanied with such a complex product. We need to convince stakeholders why accessibility matters even if there are few PWDs using our product.

Another challenge is that most of our co-workers' knowledge on web accessibility is limited to what it is, why it matters and there're certain standards and guidelines. No one has had related training before, let alone accessibility experts in this particular domain. Lack of deep understanding and expertise in development makes accessibility implementation more difficult. When we started to read through WCAG 2.1, we found that the English version is quite challenging to comprehend for non-native English speakers like us, yet the official Chinese translated version is obscure and hard to understand.

3 OUR PRACTICE

3.1 Bringing Stakeholders on Board

In order to persuade stakeholders to buy in on the accessibility initiative, we need to articulate who can benefit from this and how. Certainly, improved accessibility would bring tremendous value to our impaired users, but stakeholders need more explicit data to make informed judgement. Therefore, we conducted an online survey on our users' physical and mental condition, as well as their needs and perception on the accessibility of Alibaba Cloud Console.

The questionnaire link was posted as a message on top of the homepage of our cloud console and the landing page of 3 major products, which can only be discovered after login. Thus, participants of our survey are guaranteed to be users really using the console. The data collection duration lasted for three weeks. We collected 2675 valid responses in total. All the participants are from China, and most (76.8%) of them are aged between 21-40. Not surprisingly, only 6.7% of the participants are with disabilities. However, as shown in Figure 1, 14.28% of the participants mentioned that they have encountered one or more types of temporary disabilities or situational limitations [6]. We also asked all our participants to select from a list of accessibility related issues that they have experienced when using our cloud console, such as cannot operate via keyboard, lack of clear guidance, chart not conveying information clearly, poor content readability, etc. Up to 34.62% of the participants selected one or more issues from the list, while the rest of the participants indicated that they haven't experienced any of the suggested issues. Moreover, 38.17% of the participants expressed their needs for better accessibility, such as higher contrast and keyboard access. And among the 56.93% participants who know about accessibility, 70.7% think it's necessary to make Alibaba Cloud Console more accessible.

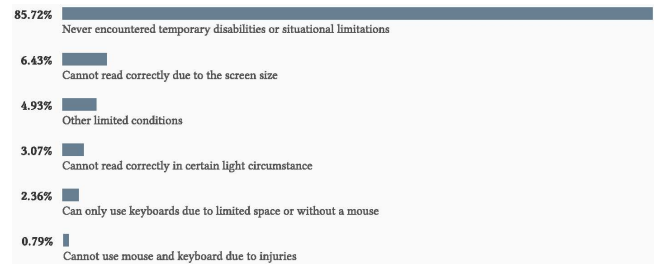


Figure 1: 85.72% of participants have never encountered temporary disabilities or situational limitations. 6.43% of total participants selected “cannot read correctly due to the screen size”. 3.07% of total participants selected “cannot read correctly in certain light circumstances”. 2.36% selected “can only use keyboards due to limited space or without a mouse”. 0.79% selected “cannot use mouse or keyboard due to injuries” (In our questionnaire, the “Never encountered temporary disabilities or situational limitations” option is mutually exclusive to others, and the other options are multi-selective.)

The survey result mentioned above makes a strong argument in terms of how many people are actually in need for better accessibility on our platform, in addition it tells a data-driven, yet convincing story of why we need to raise our awareness toward providing accessible product to our users. With that being said, what we need to accomplish next is to come up with a feasible and cost-efficient plan that will facilitate design and development.

3.2 Tailored Accessibility Guidelines

When we started out implementing accessibility in Alibaba Cloud Console, we referred to WCAG 2.1 as our implementation guidelines, as it is an internationally recognized standard. As much as the content of WCAG 2.1 is clear and well organized, with plenty of design and development techniques being provided, the dense and verbose nature of this kind of guidelines could appear to be very intimidating and hard to digest. Besides, the guidelines were originally written in English and the translated Chinese version is obscure in various aspects (for example, many terms are directly translated without any localization), which makes it harder for non-native speakers to take in. As the design team of Alibaba Cloud Console, we want to explore how we could adapt WCAG 2.1 to our own design and development process.

3.2.1 Survey & Analysis. In order to evaluate how well our designers could understand WCAG 2.1 and how much they could apply it to their design, we conducted a readability survey on WCAG 2.1 among the design teams of Alibaba Cloud Console, which consisted of 13 members (5 males and 8 females). These participants are aged from 25 to 32 years old. These participants have been responsible for the user experience of Alibaba Cloud console for 5 years averagely and at least 2 years. Therefore, they were proficient in console-related design. In the survey, each participant was presented with the official translated version of WCAG 2.1 content and was asked to record the time used for reading through

Table 1: Designers Rating on WCAG 2.1 Guidelines

Rating Aspect	1-7 Scale	85% Ratings	Average Rating
Easy to read and comprehend	tough to easy	3 or less	3.08
Easy to apply to design of Alibaba Cloud Console	tough to easy	4 or less	3.08
Relevant with design of Alibaba Cloud Console	irrelevant to relevant	4 or less	3.31

the whole guidelines. It turned out that 10 out of 13 designers spent over 30 minutes to finish reading, including 5 designers who spent more than 1 hour.

We also asked them to rate how they felt about the guidelines in terms of ease of understanding, relevance to their daily design work and ease of application. The result is shown in Table 1

The results indicated that learning and using WCAG 2.1 was quite a tough and time-consuming task for our designers. According to our survey, one of the main problems is language and expression. Though provided with the official translated version, participants still find the phrasing obscure. Another key problem is that WCAG 2.1 contains excessive information that's not relevant to design of Alibaba Cloud Console.

As our cloud console now contains over 100 products supported by more than 30 designers, it seems impossible to train every designer in a short period of time for them to get encyclopedic knowledge of WCAG 2.1 and apply to their work. Instead, we need a more adapted, simplified and comprehensible version of WCAG 2.1.

3.2.2 Our Optimization for Console Design.

Focus on relevant content. The first optimization we made was to reduce the volume of content by filtering out content that's irrelevant to our current design work. We carefully selected the content in the context of our console design. As the interfaces in Alibaba Cloud console are mostly composed of textual content and seldom involved with media content like pictures and videos, we filtered out the "1.2 Time-based Media" guideline along with 9 success criteria included. We also removed "2.3 Seizures and Physical Reactions" guideline along with its 3 success criteria included, as motion and flashing hardly occur in our console. Given the complexity of tasks that our users perform on our console, they rarely use mobile phones or small screened devices to operate on those tasks. According to the data we gathered, 92.69% of the users are using devices with resolution higher than 1280*70 pixel. Thus, we eliminated 3 success criteria related to screen, such as "1.3.4 Orientation", "1.4.10 Reflow". Therefore, our designers could devote more efforts to the more relevant guidelines.

Restructure and Rephrase. To further assist our designers in understanding the guidelines easily, we then deconstructed the original structure of WCAG 2.1, which is organized by four top principles "Perceivable", "Operable", "Understandable" and "Robust". We reconstructed the contents from a designer's point of view while taking development into consideration. The modified structure consists of 6 principles, starting from instruction on generating proper visual system, including color contrast, fonts and layouts, followed by hierarchy design, behavior design and content design, and lastly implementation guidelines for pointer and keyboard

Table 2: Comparison of structures of WCAG and ACCG

WCAG 2.1 Principles	ACCG - Principles	ACCG - Suggestions Number
Perceivable	Color and Contrast	5
Operable	Fonts and Layouts	2
Understandable	Hierarchy & Navigation	3
Robust	Behavior & Feedback	4
	Content Design	2
	Pointers, Keyboards & Assistive Technology	5

operation. It provides accessibility instructions at every phase of the product design in a sequential manner.

On top of this structure, we further reorganized the remaining 69 success criteria of WCAG into 21 actionable suggestions that designer could follow and directly apply to our console design. We also rephrased and localized them into native Chinese expression. The structure of Alibaba Cloud Console Accessibility Guidelines (ACCG) is shown in Table 2

Make it executable. To make sure the Alibaba Cloud Console Guidelines can be absorbed precisely and applied correctly to console design, we supplemented every suggestion with the actual use case of Alibaba Cloud Console as examples. For instance, for the suggestion "Don't use images to convey meaningful text information", we provided an example of common error when developing pages of Alibaba Cloud Console, so that the designers could have a clear and contextual view of the adapted guidelines.

4 RESULTS

To evaluate if the adapted guidelines stated above is actually more desirable in our own use case, we ran a second-round survey among the same participants with same questions and rating scale. The results show that the new guidelines performed better on every aspect mentioned above. Firstly, all participants were able to finish reading the guidelines within 20 minutes, among which 73% of them used no longer than 15 minutes. The rating on the three aspects above were also greatly improved. The average rating of each aspect is increased by at least 47%. The detailed comparison of the ratings is shown in Table 3

Besides the results we gained from the survey above, we also tried to collect their feelings on the adapted guidelines. We had a short discussion with each participant after they finished the survey. Every participant said that adapted version was much easier to comprehend. Many of them also mentioned that the suggestions

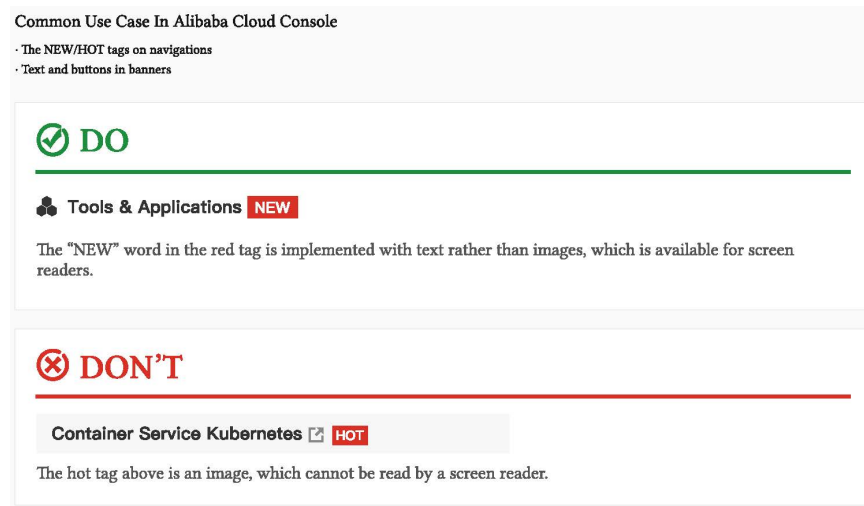


Figure 2: Example of “Don’t use image to convey text information”.

Table 3: Designers Rating on WCAG 2.1 vs ACCG

Rating Aspect	85% Ratings on WCAG	85% Ratings on ACCG	Average Rating on WCAG	Average Rating of ACCG
Easy to read and comprehend	3 or less	5 and above	3.08	4.77
Easy to apply to design of Alibaba Cloud Console	4 or less	5 and above	3.08	4.54
Relevant with design of Alibaba Cloud Console	4 or less	5 and above	3.31	5.23

and vivid cases we offered in ACCG could provide great instructions when designing for Alibaba Cloud console.

4.1 Level Redefined

In order to meet the needs of different groups and different situations, three levels of conformance are defined in WCAG 2.1: A (lowest), AA, and AAA (highest) [7]. One of the factors evaluated when setting the level is whether it is possible for all web sites and types of content to satisfy [8]. In another word, these levels of conformance are defined to be applicable to all websites, instead of targeting a specific type website that serves a specific group of users. For instance, according to our survey, only 6.7% of our users have one or more disabilities, and only 1.01% need a screen reader or other assistive technology to use our cloud console. The survey gives us a clear picture of which part of WCAG is more impactful to our current users. On the other hand, the WCAG level definition takes into consideration whether content creators could reasonably achieve the required skills. But it doesn’t consider the implementation cost for an existing complex website. Therefore, with limited implementation resource, we need to come up with a new set of level definition so that we could prioritize tasks that are most beneficial to our current users and then slowly expand to more prospect users. In our new revised version, the lower the

level is, the more users it serves and the less cost it requires. The definition is as follow:

Level 1 – Basic: This level includes guidelines that accommodate people with low vision and/or cognitive limitations (2.07%). It also covers the 14% of participants who have temporary or situational limitations, which includes users who may be viewing our console in poor lighting conditions or whose vision is worsening with age (3.4%), and users who have difficulties using our console due to limited screen sizes (6.43%). The corresponding guidelines in WCAG 2.1 are mainly “Distinguishable”, “Enough Time”, “Readable”, “Input Assistance” and “Predictable”. There are a few success criteria that relates to other guidelines like “Navigable”, “Input Modalities”, etc. Figure 2 shows the detailed success criteria included in this level.

Many of these criteria could be directly perceived, even by users without disabilities as well. For example, on the visual aspect, higher contrast is demanded by 18.73% of the participants, and larger font size and line height is wanted by 5.83% of the participants. As for cognitive issues (Figure 3), 13.35% of total participants indicate that our content and visual hierarchy is unclear, 11.81% think the content lacks clarity, and need more help and tooltip.

Besides serving a large number of users, the success criteria in this level are relatively easy to implement in our cloud console. For example, some criteria related to “Input Assistance”, “Enough Time”, “Readable”, “Input Assistance” and “Predictable” guidelines

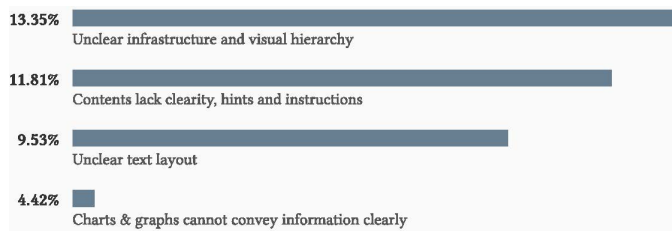


Figure 3: 4 major cognitive issues related to our console accessibility indicated by participants.

are already included in our existing design guidelines, and most of our products have already met these criteria. Thus only a few piecemeal improvements need to be done. We also renovated our UI component library to meet the criteria related to “Distinguishable”, so that all the interfaces in our cloud console built with these components were automatically upgraded to an accessible version.

Level 2 – Intermediate: This level is defined to accommodate PWDs relying on keyboard operation or assistive technologies that are operated through keyboard interface, including 2.4% participants who are temporarily and situationally relying on keyboard, as well as people who prefer to navigate and interact with a keyboard and no mouse for efficiency, which accounts for 15.2% of our users. Achieving this level means supporting full keyboard access. The success criteria (Figure 4) in this level relates to WCAG 2.1 guidelines like “Keyboard Accessible”, “Navigable” and some others. We are currently working on the implementation plan of Level 2.

Level 3 – Advanced: This level depends on realization of Level 2 and is defined to accommodate blind users who rely on using screen reader to operate. Most success criteria in this level are extracted from guidelines like “Text-alternatives” and “Compatible” in WCAG 2.1. Since our survey indicates only 1.1% users require the support of screen reader, this level will be the next step once we accomplish Level 1 and Level 2.

With the newly defined 3-level model, we broke down our goal into feasible and cost-manageable steps. And this also helped bring our stakeholders on board as they could easily see how each phase of work bring value to our users explicitly.

5 DISCUSSION

As much as limitations exist, we hope our approach of taking incremental steps toward building a more accessible product would serve as a positive reference for other enterprise companies in building more inclusive and accessible products in the future, especially for complex products with a lot of moving pieces.

Our adapted guideline is a simplified yet effective version of WCAG 2.1 tailored specifically for our enterprise scaled product. Therefore, it might not be as comprehensive as WCAG 2.1 and could possibly still have blind spots on certain edge cases when used at scale. Also, as we are focusing more on the design side currently, the development part of the guidelines is not as detailed as expected. We only tested its readability comparing to WCAG 2.1 with our designers. More studies need to be performed to see if it is also more usable for developers comparing to WCAG 2.1 or other accessibility development handbooks they may refer to, and then modifications need to be made accordingly.

Another dispute lies in how we define the importance of the 3 levels. For the 1% of our users who rely on a screen reader, our console is completely not usable if Level 3 is not achieved. While for the majority of users that Level 1 targets, our console is still usable in most circumstances. So, it’s more of a business decision to make when determining which should be prioritized.

6 CONCLUSION AND FUTURE WORKS

Incorporating accessibility into an existing product with such complexity is a cumbersome yet challenging undertaking with many things needed to be taken into consideration. Our practice provides a feasible example of how to take baby steps to make it more executable. The survey shed light on how our users view accessibility in their everyday workflow and further helped us with establishing the revised level definition to determine the priority of work order. The tailored guideline made accessibility more consumable by our designers as it’s more customized and compatible with our cloud console. Our guidelines and level definitions may not be directly usable for fellow enterprises, but we hope the approach itself may serve as a reference.

Although most of Level 1 success criteria can now be met by upgrading to the newest accessible version of our UI components, there are still some cognitive related issues that we need to assess and fix across our console. We are also working with our developers to compile a more development-oriented version on top of our tailored guidelines, so that developers of different products in our console can implement accessibility in the code with ease. Reducing the learning curve for developers is essential to achieve Level 2 and Level 3. Based on the three levels that we defined, our next step will include establishing an evaluation method to evaluate each product in our cloud console.

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